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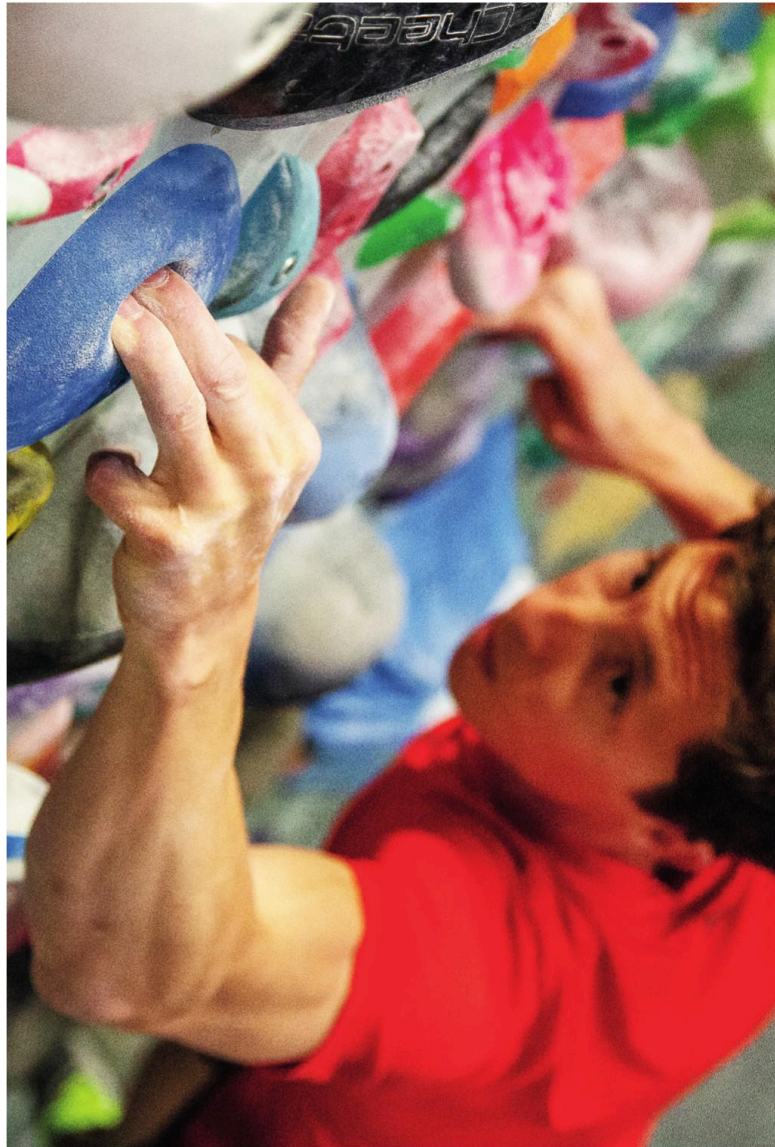
THE INJURIES



INJURIES AND BODY PARTS

For each part of the body, the most common injuries and ailments are presented, and for each injury/ailment, the presentation of the injury, diagnosis and management are discussed.

No matter how careful we might be with sleep, nutrition, load management and strength training, injuries are undeniably a part of sports. Even though we can greatly reduce the risk through the measures described in the previous chapter, unfortunately sometimes 'shit just happens'. Just think about the forces our fingers are exposed to when we climb. A large proportion of our body weight is supported on just a few millimetres of grip surface while we move our centre of gravity towards the next hold. The combination of the force development required to maintain a grip position and all the small movements that occur in the finger joints while moving the body can easily exceed what the different structures in the fingers can handle. But this is a risk we take every time we climb. In our pursuit of progress, we will always be balancing on a knife edge between training hard enough and training too much.



It's important to emphasise that the prognosis – how things will turn out – for the injuries presented in this book is generally good. Most likely, if you pick up one of these injuries, you will fully recover! First and foremost, you need to get an accurate diagnosis so that you can quickly engage in a plan for full recovery. This plan should include specific measures for the injured structure, but also more general ones. After all, a finger injury only concerns one small, albeit very important, body part involved in climbing. Training technique, tactics, and overall strength and conditioning will not only contribute to faster healing, but also make your time off due to injury more fulfilling and help you become a better climber. Remember this as we delve into the dark material regarding the potential injuries that climbers may face.

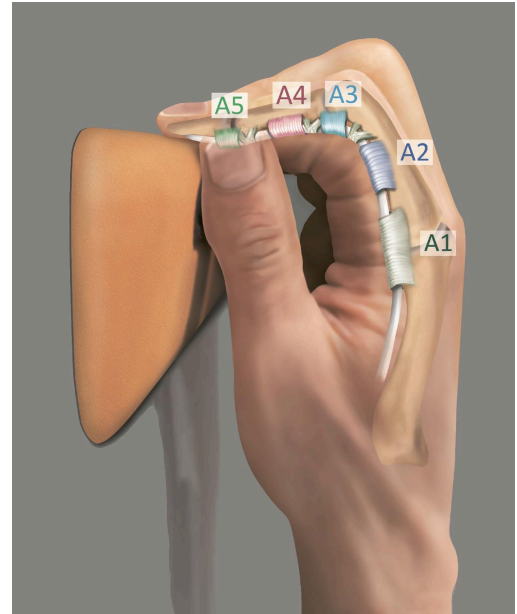
FINGERS: PULLEY INJURIES

'Pop.' The sound itself is difficult to describe. Perhaps it was more of a feeling? Like something slipping or tearing. But those around me wonder if it was my joint that made the characteristic cracking sound that one can make oneself, so there must have been a sound. I know what this sound means. I've lost count of how many people have described this sound in my clinic. As I stand on the mat and press around my finger where I know the A4 pulley should be, it occurs to me that this doesn't hurt at all. But when I put a little more pressure on my finger, I notice that it's actually quite painful to half crimp, and any remaining hope I have of climbing again today disappears. Inside my head, I've already started forming a plan. A plan for how to get back to doing what I love as quickly as possible. This will be fine. Shit happens. This will be fine.

This is my own experience, but it could apply to many others since pulley injuries are by far the most common finger injury in climbing. And the descriptions are quite similar: a palpable and/or audible pop in a finger, followed by varying degrees of pain when pressing on or loading the finger. In the days following the injury, swelling may occur, along with reduced mobility in the finger and decreased strength due to pain during loading.

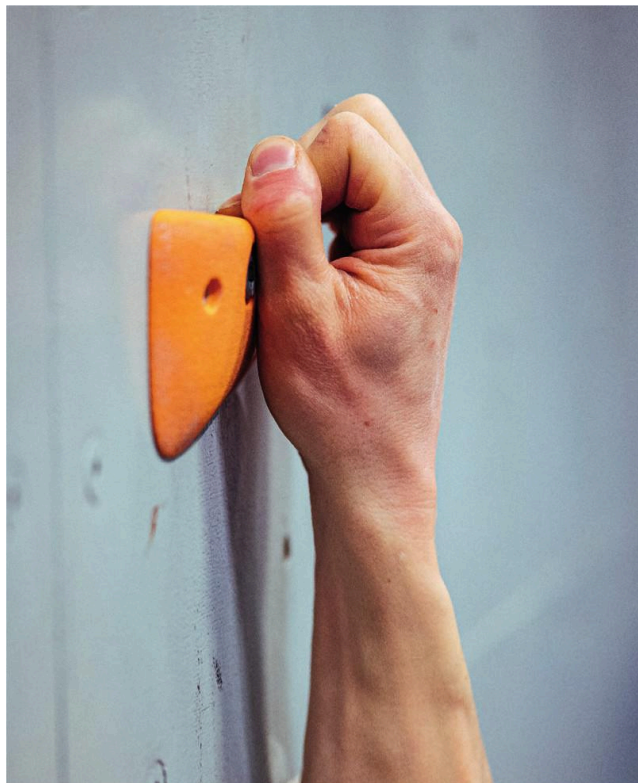
Each finger has five annular pulleys – except for the thumbs, which have two pulleys each – and they are reinforcements of a larger layer of connective tissue covering the palm and the inside of the fingers. The role of the pulleys is to keep the flexor tendons close against the bones so that the bending of joints becomes possible, much like the rings on a fishing rod allow it to bend.

The finger pulleys are numbered from innermost (A1) to outermost (A5). The stiffest and strongest pulleys are the A2 and A4, and these are also most prone to rupture.



CRIMPING, FRICTION, FORCE AND INJURY

Crimping involves bending the middle (proximal interphalangeal, or PIP) joint of the finger by more than 90 degrees while stretching the outermost (distal interphalangeal, or DIP) joint. The thumb is often placed over the index finger as well. The crimp grip is many climbers' favourite grip position on small holds, yet this hand position is closely associated with pulley injuries – so why do we use it?



We use it because it offers several advantages:

- we can involve an extra finger (the thumb) to generate force;
- we can move our centre of gravity closer to the next hold when compared with an open-handed grip position;
- and we're improving (lengthening) the lever arm so the flexor muscles can apply more force to the grip surface.

In addition to all of this, the insides of the A2 pulleys are lined with small grooves that match the surfaces of the superficial flexor tendons. This allows the surfaces to lock against each other, like a Chinese finger trap, when the friction between the structures increases. Due to how these grooves are organised in relation to one other, tendons can slide relatively unhindered inwards – when the fingers bend – but encounter resistance outwards, both during static holds and when fingers straighten under load. This locking mechanism – called *tendon locking mechanism* – is what allows bats, birds and certain climbing mammals to hang from their fingers and toes without using muscle power to maintain grip.

By measuring differences in force development when actively trying to flex fingers against resistance versus hanging still, we see that this system can bear more weight while hanging. The difference lies in the friction created between tendons and pulleys. Single studies have shown that friction accounts for up to nine per cent of force development at mid-joint level in a finger. This means that by crimping while hanging, either nine per cent less muscular effort is required, or nine per cent more force can be exerted on a hold during maximum muscular work. Friction between tendons and pulleys is believed to play a role in pulley injuries since these injuries often occur due to a sudden increase in load that causes the fingers to straighten slightly – eccentric loading – thereby increasing the degree of friction.

The tendon locking mechanism also directly affects how we should train finger strength. To influence pulleys so they can withstand higher loads, it is actually beneficial to train a controlled crimp grip. On a fingerboard, we can easily adjust both weight and duration while being certain that unexpected increases in load should not occur. This training will over time lead to thicker and stiffer pulleys, and will also reinforce the correct use of the crimp grip position. And this is important because when climbing on small holds close to our maximum level, we'll almost certainly need to crimp.



By using a grip tool and a load cell (such as a Tindeq) fixed to the ground, we can accurately measure how much force we are able to produce by curling our fingers against the grip. There is no movement, so the muscle contraction is isometric, but the load cell registers how much force we are able to produce against solid resistance, without the extra help we get from the locking mechanism in the connective tissues.

On the other hand, we try to avoid using the locking mechanism for optimal muscular effect during training. The method of *overcoming isometrics* has been used in strength training for several years, but it is only recently that climbers have realised its potential. With portable force sensors/cells and fingerboards, it's easy for us to curl our fingers against fixed resistance and accurately measure force development (see [photo above](#)). A less technological variant involves performing similar movements with our fingers on a fingerboard while standing on bathroom scales and seeing how many kilograms we can subtract from our body weight. Since we avoid the locking mechanism in both cases, we can train to increase our finger strength without putting too much strain on the connective tissues.

MECHANISM OF INJURY AND DIAGNOSIS

When a finger is bent, the flexor tendon is pushed against the pulley system. If the load becomes too great, the pulley may rupture partially or completely – referred to as partial and total ruptures. In unfortunate cases, multiple pulleys can rupture simultaneously, known as multiple ruptures.

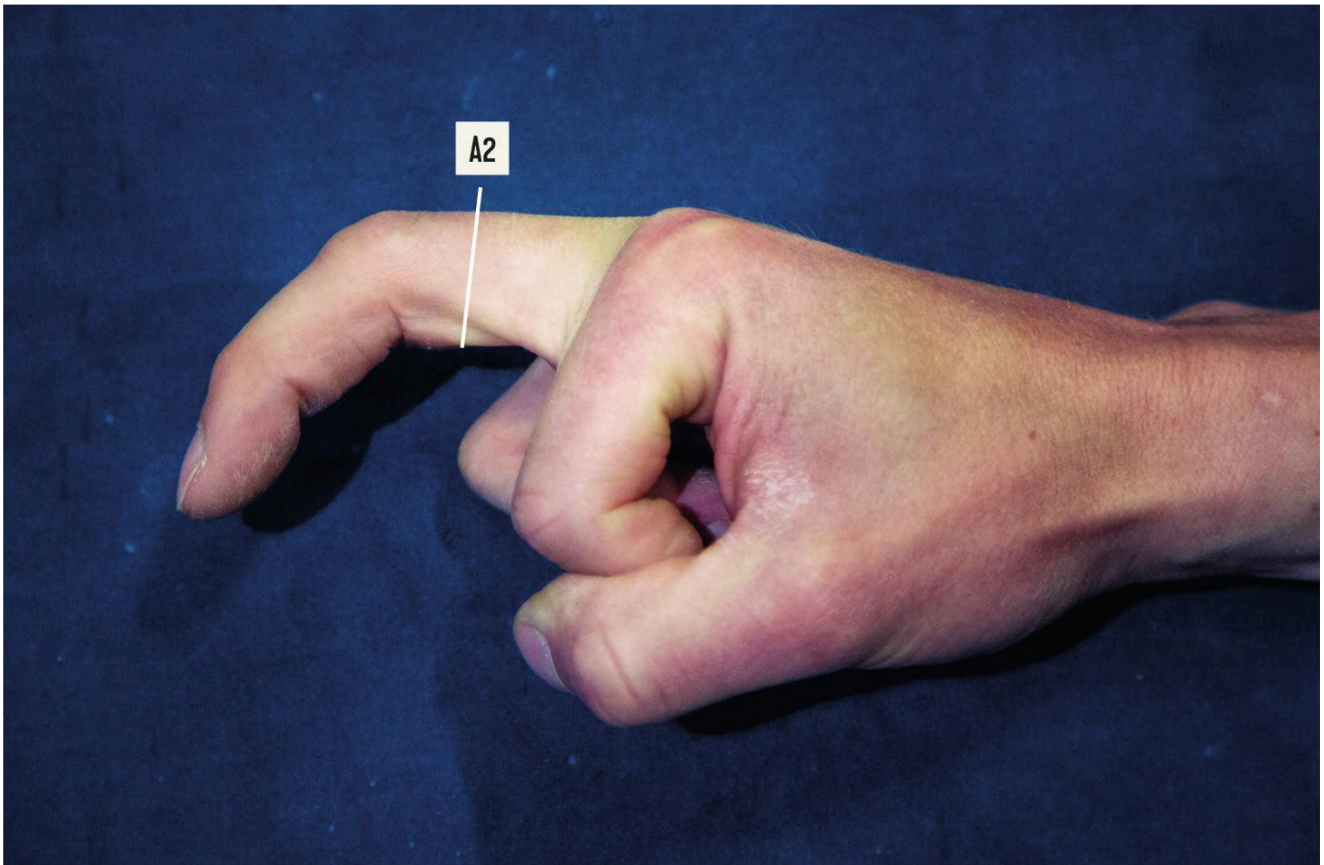
The most common ruptures are to either the A2 or A4 pulleys, and they are most often caused by a combination of small hold, crimp grip and a sudden increase in load. For example, this could happen if a foot slips while you are crimping hard, or while you are transitioning from an open-hand grip to a crimp grip. Rupturing a pulley sounds and feels more dramatic than it actually is. In many cases, there may be an audible

‘pop’ when the pulley ruptures, followed by pain upon loading of the affected finger. Swelling will gradually occur around the torn pulley, causing stiffness when bending the affected finger.

It is advisable to have an examination to confirm that it is indeed a pulley injury and to assess its extent. In most cases, a clinical examination supplemented with ultrasound imaging should suffice.



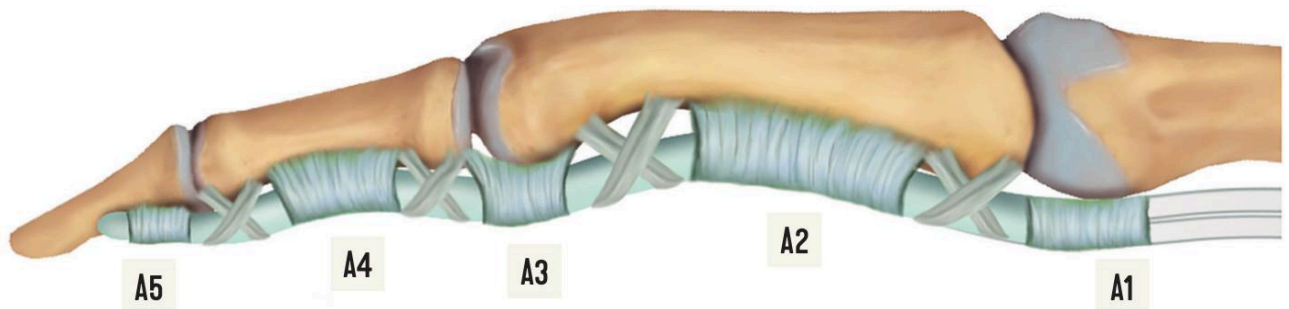
Lay your hand flat with your palm facing up. Bend at the middle joint – the PIP joint – and press the fingertip against resistance, such as another finger. Pain over the A2 or A4 when the finger is stressed in this way may indicate a pulley injury, but this should also be consistent with an acute event that triggered the injury.



MANAGEMENT

Even in larger injuries involving two or three pulleys, the focus is on rehabilitation, but it is important to determine the extent of the injury in order to tailor this rehabilitation as best as possible. German doctor and climber Volker Schöffl has proposed a classification and rehabilitation protocol for different types of pulley injuries, but, in my opinion, this is relatively conservative and there are significant individual differences in how quickly you can progress with rehabilitation. For example, an A2 rupture may require a slightly slower progression than an A4 rupture, and two climbers with A2 ruptures may also have unique rehabilitation processes. Currently, no one has the definitive answer to how to rehabilitate pulley injuries optimally. The key questions revolve around how soon we start loading and how quickly we proceed with different types of loads.

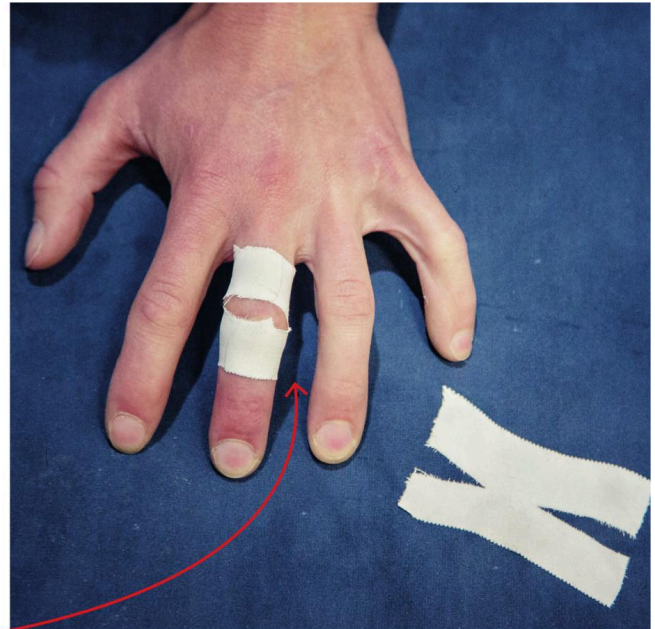
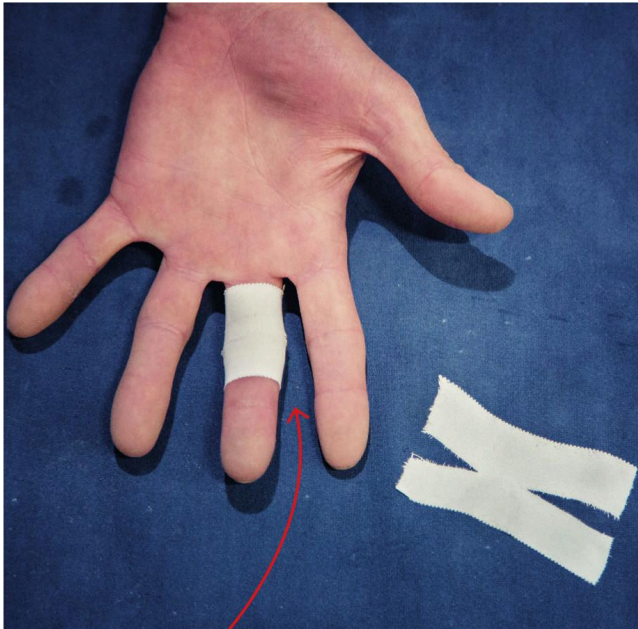
Once a pulley injury has been diagnosed, it is appropriate to give the affected finger a few days of rest. However, rest does not mean keeping it completely still; it is important to move the finger even if you do not exert much strain on it. Although the ligament itself will not heal back to its original form because of the formation of scar tissue, it is important to create tension in this scar tissue so that it shapes itself as best as possible for future load bearing. In cases of multiple ruptures, however, using a splint – a so-called pulley ring – is important for stabilising the flexor tendons against the bones. This way, over time, scar tissue will adhere more closely to the bones and take over the function that was previously performed by the pulleys. As mentioned earlier, pulleys enable the bending of finger joints; if tendons are not held against bones and instead bowstring from the palm to fingertip, this will result in loss of function while also leading to clawing fingers.



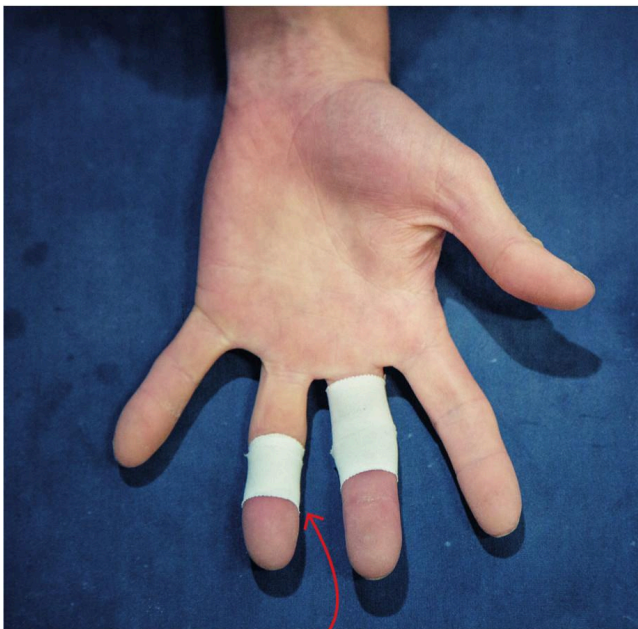
The use of pulley rings is also recommended for total A2 ruptures based on the same principle of stabilising the tendon against the bones. However, only one study has been conducted on this approach, and although the results were good, they have not been compared to a control group that did not use a splint. Therefore, it is difficult to say whether this approach is better than not using a splint and whether it is necessary for achieving good short- and long-term outcomes. It is important to note that isolated A2 ruptures have only recently been treated with splints, and we know that this injury has a good prognosis without negative long-term effects even without using a splint.

Whether a pulley ring is used in the first few weeks before loading or if loading is started early on, the injured finger must be loaded. As mentioned earlier, it is through loading that scar tissue settles down in the best possible way and provides both tendons and connective tissues with the elasticity, stiffness and capacity needed for climbing again at your desired level. Naturally, this load also increases the muscular force development you are completely dependent on. By using fingerboards or grip tools with weights or force sensors/cells, you can maintain control over weight, time under tension, grip position and tempo.

Loading an injured finger will eventually cause pain, but as long as it's not so painful that it becomes impossible to handle, you can challenge it. The most important thing is realising that rehabilitating an injured finger cannot be achieved by training pain free, and the pain felt during training is not dangerous. If you load too heavily, you won't be able to generate any power because it will hurt too much, resulting in ineffective training. If you progress too quickly, you may experience increased swelling after training which is also counterproductive as it disrupts continuity in training. Therefore, you must aim to find a 'sweet spot' between load and symptom response, focusing primarily on potential increased swelling after training. The first priority is to gradually increase the load, and you should usually start with open grip positions on relatively good holds (>20-millimetre-deep grip surface) before gradually progressing to half-crimp grips and smaller holds. When the pain reaches a point where increasing weight becomes impossible, you can make the training more challenging by increasing hang time to fatigue the muscles. However, it is important to be able to increase load during rehabilitation, and as load increases, hang time decreases.



H-taping an A2 injury.



Circular taping an A4 injury.

In the first few months after a pulley injury, it is recommended that the finger is taped to relieve stress. H-taping is recommended for A2 injuries (see photos, top left and top right) and circular taping is recommended for A4 injuries (see photos, bottom left and bottom right). The tape should be applied tightly to provide the best possible support, but be careful to maintain blood flow to the finger. As you gradually put more load through the finger, the tape will no longer have the same supportive function and there is no need to tape a previous pulley injury once you have returned to full weight bearing.

During rehabilitation, climbing is important for maintaining movement skills, and initially, it will be crucial to control movements and grip positions on the wall. As you feel confident in increasing load in the specific training, you can also challenge yourself more on the wall, but here the focus should still be on experiencing progress with the finger rather than being able to complete a boulder problem or route. You should gradually work towards developing as much force with the injured finger as before the injury, and this will involve being able to develop this force quickly. This can be done through jump-starting or kick-starting to campus rungs or different holds, where the key is to hold the position on the grip or hold you have jumped to. For example, you can start with a short distance – i.e. less speed in the movement – on good campus board rungs, and then progress to increasing the distance, using smaller holds and different hold types on various wall angles.

This progression ladder – starting with as much weight as you can tolerate and slightly longer hang times, then increasing the weight and reducing hang time, and finally working on rate of force development – will enable you to gradually return to your previous level. And there is no reason to stop there. By continuing this

training, you will gradually become stronger than before the injury, thus turning a period of injury into something positive.

There is no definitive answer as to how long the process will take before you are back at your previous level, and from experience I have seen significant individual differences. However, from a purely biological perspective, it makes sense to estimate 8 to 12 weeks before the tissue allows for a load that approaches where you were before the injury. Another crucial factor is how confident you are when loading the finger. Some people have no problems with this, while others are naturally more reserved. This adds another dimension to rehabilitation. It's not just about loading in order for the tissue to withstand stress, but also about loading in different ways to regain confidence in pushing yourself as much as possible. The time it takes for this can vary greatly, but from experience many people are physically and mentally ready to return to their previous level three to four months after an A2 pulley injury.

And if you do rupture a pulley ... don't worry! There is nothing about an A2 pulley injury that suggests you won't reach your desired climbing level.

SUGGESTED REHABILITATION: PHASE 1

Load the tissue early in a controlled way.

Avoid an increase in pain and swelling during and after training. As the pain subsides, you can load heavier; if there is no subsequent swelling, you can move on to subsequent phases.

GRIP POSITION:

Four-finger half crimp on a 20mm edge and/or an open grip position on a sloping edge.

METHOD:

No-hang on a fingerboard or with a grip tool, adding weight or using a load cell.

DOSAGE:

15 seconds' hang time with an acceptable level of pain; 5 sets performed daily.

CLIMBING:

On shallower wall angles (<30 degrees), using open grip positions. Focus on footwork, balance, weight transfer and clipping technique.



NO-HANG

By standing on the floor and pressing your fingertips against a hold/grip tool (such as a fingerboard), you can load the injured finger with as much weight as you can tolerate in terms of pain. This is a simple and low-tech way to get started with weight bearing, but it will not show you how much you can actually load the finger. You could therefore stand on bathroom scales and see how many kilos you can take off your body weight.



LOAD CELL

The best way to load a finger is to use a grip tool with a load cell and perform 'overcoming isometrics' as explained on page [43](#). This will give you an accurate measure of how much force you can develop before it becomes too painful to continue. In this way, you can measure progress in terms of increased strength development and manage progress according to symptom response during and after training.

SUGGESTED REHABILITATION: PHASE 2

Gradual increase in load on a fingerboard or with a grip tool.

GRIP POSITION:

Four-finger half crimp on a 15–20mm edge.

METHOD:

Deadhangs or overcoming isometrics using a grip tool with weights or a load cell. No-hangs from phase 1 as warm-up and part of the daily routine.

DOSAGE:

Fingerboard/grip tool: 5–10 seconds' hang time; 5 sets, 3 times a week.

NO-HANG:

10 seconds x10 with <50% body weight.

CLIMBING:

Controlled climbing using smaller edges on wall angles <30 degrees, and on a wider variety of open grip types on steeper terrain. Avoid incut jugs as these will compress the A2 and A4 regions of the finger.



DEADHANGING

In this phase you can introduce hanging exercises in addition to continuing the training from the previous phase. Start with relatively good holds and lift your legs off the floor so that you can hang freely from your fingers.

SUGGESTED REHABILITATION: PHASE 3

Increased weight, increased rate of force development and increased demands on the finger during climbing.

GRIP POSITION:

Half crimp on a 6–20mm edge. Full crimp during no-hangs to re-introduce the grip position in a controlled way.

METHOD:

Deadhangs or overcoming isometrics using a grip tool with weights or a load cell. No-hangs from phase 1 as warm-up and part of the daily routine. Kick-start/jump-start to a variety of grip types, gradually increasing distance and reducing hold size.

DOSAGE:

Deadhangs/grip tool: 3–5 seconds' hang time, focusing on rate of force development. Perform as repeaters: 3–5 seconds' hang time x4 with 5 seconds' rest between reps; 4 sets, twice a week. Consider the need for daily no-hangs.

KICK-START/JUMP-START:

4 jumps; 2–3 sets, 2–4 times a week.

CLIMBING:

Specific training on smaller edges on steeper terrain, working on single moves to and from half-crimp and full-crimp grips.

An acceptable level of pain that does not hinder force development while loading is okay. Worsening of pain during exercise indicates excessive load. Increased swelling the following day is also a response to excessive load. The basic principle is frequent short sessions, with gradually increasing load and decreasing hold depth.



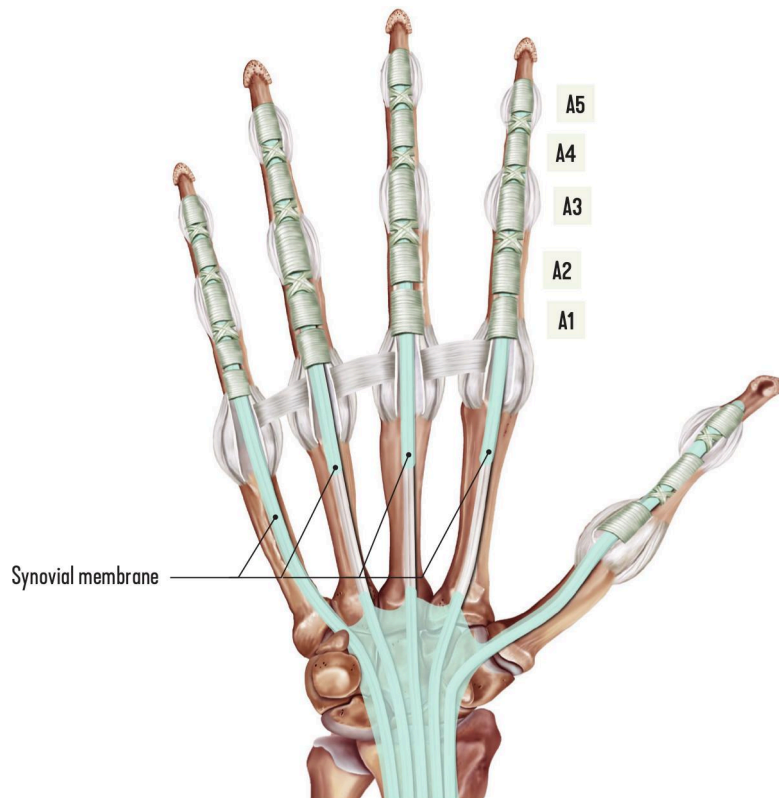
JUMP-START

By introducing jump-starts into rehab, you can stimulate rate of force development for the flexor muscles of the fingers and increased stiffness in the connective tissues. This is often perceived as scary following a pulley injury, so this exercise is a good way to build confidence in latching a hold in the half-crimp grip.

FINGERS: TENOSYNOVITIS

'I wonder if I've ruptured my pulley. It hurts here, where I've read the A2 pulley is, but I can't remember an acute incident and I didn't hear or feel anything pop in my finger. Now the finger is swollen, it hurts to crimp and it hurts to press directly on the area.'

The flexor tendons of the fingers lie inside a tendon sheath, which creates a tunnel through which the tendons can slide friction-free. The tendon sheath consists of a membrane – which is called the synovial membrane – and a stronger fibrous layer above this. The synovial membrane produces synovial fluid, and, together with the annular pulleys, this complex holds the tendons close to the finger bones and provides them with nourishment. In cases of irritation or inflammation of the synovial membrane, the production of synovial fluid increases, leading to swelling within the tendon sheath. This is called *tenosynovitis*, and it is one of the most common conditions I see in clinical practice with climbers. The condition results in a swollen finger that is painful to load and touch, and sometimes one can feel a hard lump over the area in question. This is synovial fluid that has leaked out through a small tear in the synovial membrane and encapsulated itself on the upper side of the tendon sheath. This is called a ganglion cyst.



The cause of tendosynovitis is usually climbing or training too much, either over time or in a single session. The pain can come on gradually, or more acutely after a session where you've worked a lot on hard, finger-heavy boulders or single movements. A lot of crimping is often a contributing factor, as this grip position causes the flexor tendons to push out against the pulleys and pinch the tendon sheath in the middle. A lesser-known cause is climbing a lot on incuts – very positive holds where the edge/lip of the hold presses directly against the tendons and tendon sheath (see page [57](#)). So, both when crimping and when climbing on incuts, the tendon sheath is exposed to compression forces and increased friction between the tendons and the pulleys, and this can cause an inflammatory response in the synovial membrane.

The most important response is to adjust the load and vary grip types. Easier climbing is not necessarily the best option, as many easier climbs – especially indoors – involve climbing on incut jugs and this can further irritate the tendon sheath and pulleys. The solution instead is to climb on smaller or more open holds. By climbing on less steep wall angles you will be able to reduce the load by getting more help from your legs, and by using this approach you can focus on more technically demanding boulders or routes that put less stress on your fingers. If pain permits, you can also climb harder boulders with smaller holds, but be conscious of keeping the sessions shorter and reducing the number of moves per session. I usually recommend to at least halve the session time or session volume, and slowly progress if symptoms don't increase the following day. As with pulley injuries, it is easiest to manage the load by using a fingerboard or a grip tool. You will then be able to use holds and grip positions that don't provoke symptoms, and gradually increase the load. In my experience, the symptoms improve significantly during the warm-up and then return the day after climbing. So, don't be fooled by the fact that it feels better during the session; keep the sessions short and be selective about the choice of hold types and grip positions. If there is no increase in symptoms in the form of increased pain or swelling the day after training, you have found a good balance between training dose and symptom response. The condition will gradually improve, and few need further measures such as a cortisone injection into the tendon sheath.

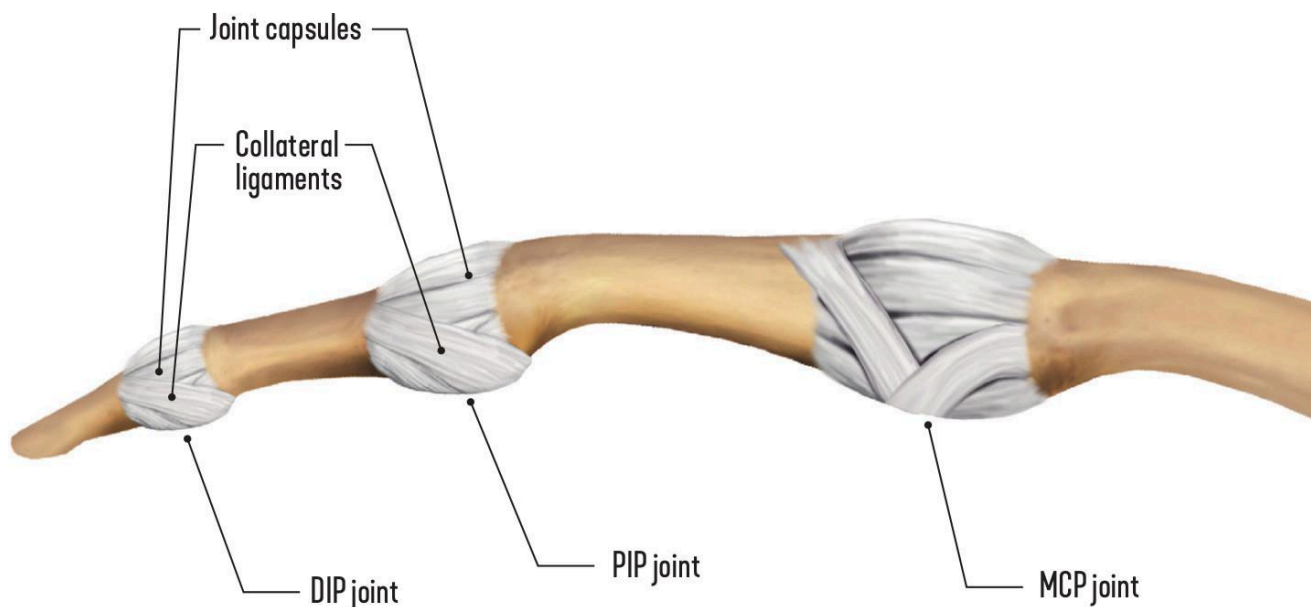
In recent years, I have recommended patients with tendosynovitis try a simple no-hang protocol once or twice a day, where the goal is to get the synovial fluid to circulate in the tendon sheath and reduce the swelling. The results so far have been good, and it is a simple measure that I recommend alongside managing the load, as described above, and avoiding holds that press directly over the affected area.



Press your fingers against the grip surface so that you take a few kilos of your body weight. Hold for 10 seconds, rest for 10 seconds, and perform 10 repetitions. Do this once or twice a day.

FINGER JOINTS

'It's here, around the joint in the middle of my finger. It hurts to bend my finger all the way in, and it seems a bit swollen after I've been climbing. I haven't done anything specific lately, but I've been climbing more frequently, and on more fingery routes and boulders.'

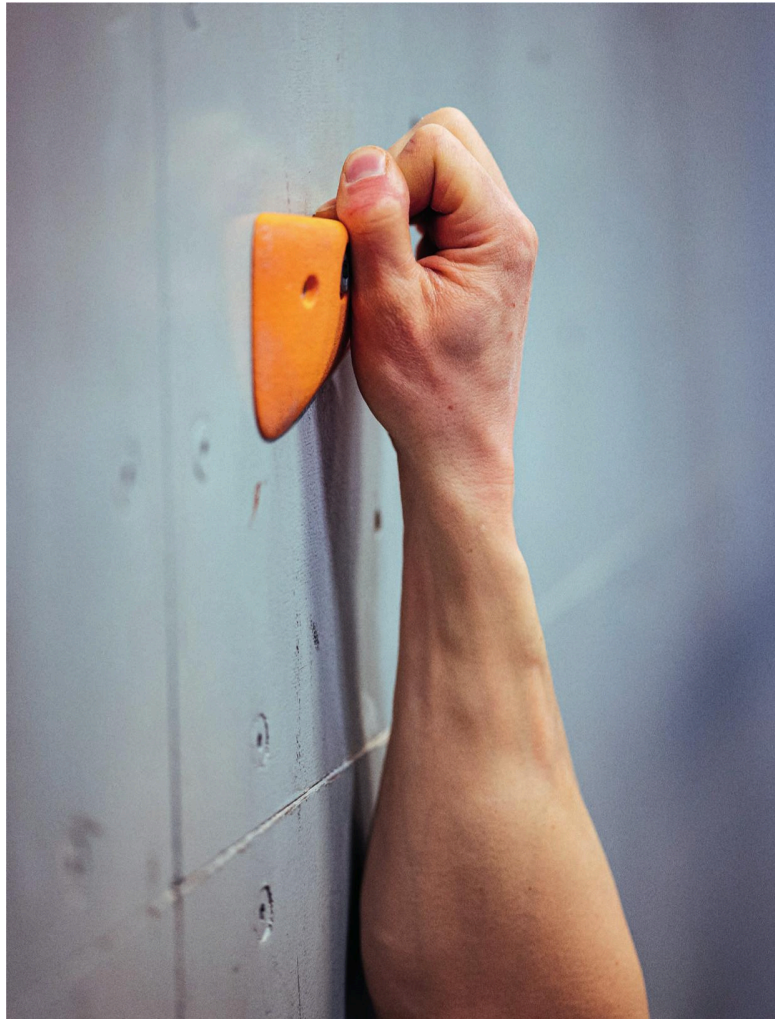


Joint-related issues in the middle and outer finger joints – the PIP (proximal interphalangeal) joints and DIP (distal interphalangeal) joints – are some of the most common reasons why climbers attend my clinic. And it's perhaps not too surprising when we think about it, because when climbing we subject these small joints to high forces on a regular basis. Although we can only actively flex and extend these joints, they can passively rotate and deviate to some extent. The combination of weight and rotational forces that these joints are exposed to can then become too much – both acutely and over time.

MECHANISM OF INJURY

Imagine yourself crimping: the PIP joint is bent to more than 90 degrees, while the DIP joint is *hyperextended*.^{*} Now imagine that the hold you're crimping isn't a horizontal edge but rather an angled grip with an uneven surface. This will subject your finger joints to forces that both rotate and laterally bend them. Thanks to the joint capsules and ligaments surrounding these joints, they will stay in place as long as the forces aren't too great. However, even if the ligaments remain intact, these forces can still cause strains and subsequently reactions in the synovial membrane lining inside a joint capsule. As a result of excessive load over time, or due to an acute incident, inflammation in this synovial membrane – synovitis – leads to increased fluid in the joint space, along with less elastic properties in its capsule. These reactions will make the joint stiffer, and bending and extending the joint will be more difficult; they will also contribute to the experience of pain when loading and moving the joint.

^{*} Hyperextension means that the joint is overstretched.



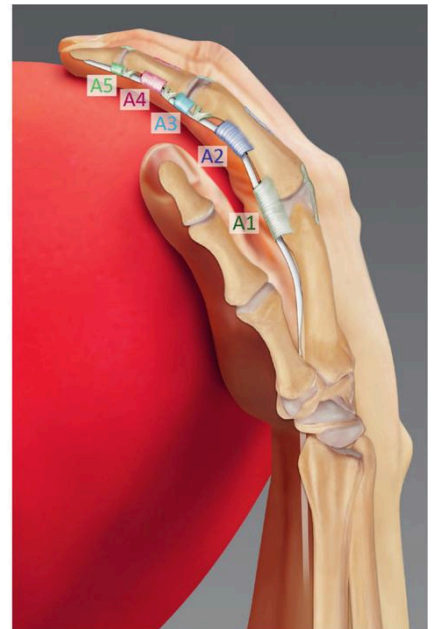
Crimping involves an acute angle in the PIP joints, hyperextended DIP joints and the thumb placed over the index finger.

MANAGEMENT

It is important to emphasise that if joint pain is related to an acute incident, such as a strain or twist injury from a fall, or from getting a finger caught in a crack or pocket, the stability of the joint should be examined by a healthcare professional. If the collateral ligaments or the joint capsule partially or completely tear because of trauma, the joint must be stabilised with tape or an orthosis (support brace) so that the injury can heal and stability can be restored.

If the joint is not unstable, management will primarily involve load management and awareness of grip positions. By using more open grip positions on less steep wall angles, the load on both the PIP and DIP joints will be significantly reduced compared to steeper climbing with half-crimp and full-crimp grip positions. This allows you to reduce physical intensity while still climbing technically challenging routes and boulders, thereby controlling finger load and instead developing other qualities that can make you a better climber. You can also control the training volume by ending sessions earlier or climbing fewer moves per session, and experience has shown that reducing volume often matters more than reducing intensity.

I often hear climbers say, *'I've been climbing easy routes, but lots of them.'* Paradoxically, this tends to worsen symptoms for two main reasons. The first is that easy routes may not actually be that easy. Even though the grades may be easy, there can still be significant physical stress on your fingers. Imagine an easy route on a steep wall. Your fingers must still bear much of your body weight, and even though the holds are good, it's relatively heavy work. And you will do a lot of it if you climb many routes in one session.



In open grip positions, the DIP and PIP joints are subjected to far less stress than in half-crimp and full-crimp positions. Varying hold selection and grip positions is therefore an important strategy for climbing without provoking symptoms in the finger joints.



Incut jugs on steeper walls are challenging in several ways. For the PIP joints, this type of hold often leads to the joints being rotated when the weight is moved from the hold. They also provide direct compression against the tendon sheaths in which the flexor tendons lie, thus contributing to the development or maintenance of a condition of inflammation in the tendon sheaths known as tenosynovitis (see pages [52–53](#)).

The second reason is that easier routes or boulders on vertical or steeper walls often have incut holds where the edges of the holds tend to press against the PIP or DIP joints.

Therefore, my recommendation is to initially reduce both volume and intensity when climbing, so that swelling decreases and pain is relieved. Measures such as icing, compression bandages and manual mobilisation of finger joints can be used, with the goal of reducing inflammation, decreasing swelling and increasing mobility. However, at present there are no studies on climbers which demonstrate that these

measures have better effects than simply managing load and using the finger in normal day-to-day activities. Nevertheless, these measures have proven effective for synovitis in other joints in the body. If inflammation cannot be reduced with these measures, NSAIDs such as Voltarol and Diclofenac in cream or pill form, as well as cortisone injections directly into the joint, may become relevant interventions.

Deadhangs are a good way to load the fingers under controlled conditions; by keeping volume per session low and hanging in varied grip positions, finger strength can be maintained despite a period of reduced training load. Pain-free training is not a prerequisite, but an increase in swelling afterwards should be avoided; both pain and swelling should return to baseline within 24 hours. This way, you can do several short finger training sessions per week, which could look like this:

- Deadhangs with varied grip positions: 10 seconds on followed by 30 seconds' rest, x5–10 repetitions.

The same principle of avoiding increased swelling will also apply to climbing sessions. As mentioned earlier, focus on gentler wall angles and open grip positions, but once the symptoms are under control, it's natural to resume more finger-intensive training. A key factor will be to keep these sessions short and stop before fatigue sets in.

Looking at the long-term changes that occur in the finger joints of experienced, high-level climbers, it becomes clear that the joints are subjected to high forces over time. These changes involve thicker bones, thicker and stiffer connective tissues, and the formation of osteophytes (bone spurs). However, these changes have not yet been linked to increased pain or reduced activity levels. Joint-related issues function in many ways as a navigational tool for managing load. Handled correctly, you can train around them when they arise, and with awareness of training load and grip positions you can reduce the risk of them occurring in the first place.

DEADHANG SESSION

10 SECONDS' HANG/30 SECONDS' REST, X5–10 REPETITIONS



Short hang sessions stimulate the joints while allowing you to build strength in a controlled environment without provoking symptoms; you can vary between different grip positions and choose a weight that suits you. There's nothing wrong with standing with your legs on the floor and doing this as a no-hang protocol if you want to reduce the load even further.

FINGERS: GROWTH PLATE INJURIES

There is one joint-related finger injury that we must handle in a significantly stricter manner: growth plate injuries in younger climbers.

BUT, FIRST AND FOREMOST: WHAT IS A GROWTH PLATE?

In order for our bones to grow in length, they have a growth plate consisting of cartilage tissue at each end. The bones grow in length as the cartilage in the central part of the growth plate expands. Eventually, the new cartilage ossifies and becomes bone tissue, and eventually the growth plate itself also ossifies. The bone is then fully grown but will continue to mature and become thicker and stronger through regular loading.

Until the growth plate is closed and fully ossified, it is the weakest link in the bone-tendon- muscle chain, with the highest risk of injury being during periods of rapid growth in both height and weight. The age at which

this period – referred to as *peak height velocity* (PHV) – occurs varies individually, as does when one finishes growing. Most growth plates are fully developed between the ages of 16 and 20, with growth primarily occurring from bottom to top within the body. Therefore, bones in the wrists, hands and fingers are among some of the last to finish growing, which poses challenges for young climbers since their fingers' growth plates are assumed not to be fully ossified until around the ages of 17 to 19.

The majority of adolescents with growth plate injuries are aged 13 to 15, which correlates with when their growth plates begin closing. This closure happens from the inside out, meaning that while the part of the growth plate on the volar side of a finger closes, the part on the dorsal side is still open. This changes how forces distribute through the bone, resulting in injuries on the dorsal aspect where the growth plate has not yet fully ossified. Injuries most commonly occur in the middle (PIP) joint of the middle finger, and are classified as Salter–Harris fractures I–V. The different numbers describe both fracture type and severity, and the most common fracture type in young climbers is Salter–Harris III.

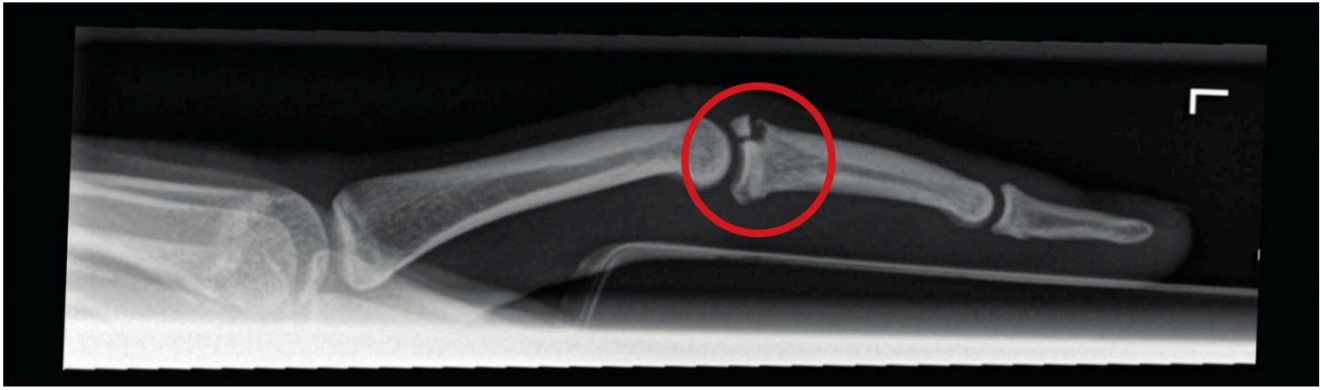
MECHANISM OF INJURY

During crimping, bouldering, campus training and weighted pull-up training on a bar, the growth plates at the PIP joints are exposed to forces that can lead to bone stress reactions, and ultimately fractures, if the load becomes too great. During their early teens, many young climbers increase their training volume and climbing level, while also experiencing rapid growth in both height and mass. It is therefore easy to understand why the risk of injury is high – it is a perfect storm of various factors occurring simultaneously. Awareness of this risk, in the young climbers themselves, and in their coaches and parents, is crucial for adjusting the load young climbers place on their fingers during puberty, and thus reducing their risk of injury.

In some cases, the injury occurs acutely with a noticeable sensation in the finger during climbing. However, in most cases there will be gradually increasing pain and swelling around the PIP joint. Range of motion will be reduced due to swelling and pain, making it painful to fully flex the finger (see [photo below](#)). Since these symptoms may develop gradually, and may not always be intense enough during climbing to hinder performance, it is important to recognise them after each session. If you are a young climber and are experiencing such symptoms, it is recommended that you stop climbing and have your finger examined.



PIP joint provocation test. By pushing the PIP joint into full flexion, it is possible to provoke pain in the event of a growth plate injury. If this is painful, I recommend you take a week off climbing and see if it gets better. If it doesn't improve after a week's rest, I recommend that you have your finger examined more closely to assess the extent of the injury and then make a plan for what to do next.



This X-ray shows a Salter–Harris type III injury with a fracture through the growth plate at the PIP joint. By resting the finger for six to eight weeks, the fracture will heal in most cases and the finger will grow back together normally.

MANAGEMENT

When it comes to pain and swelling associated with the PIP joint in younger climbers, there is a high probability that the growth plate is affected. Studies have shown that young climbers experiencing pain and swelling in the PIP joint have a 90 per cent likelihood of having a stress reaction in the growth plate. However, a clinical examination by a healthcare professional familiar with these injuries should be supplemented with an X-ray or MRI scan to assess the severity. If a fracture to the growth plate is observed, the recommended course of action is to rest the finger for six to eight weeks before taking another MRI scan to determine if the fracture has healed.

In rarer cases where there is a displaced fracture – meaning a part of the growth plate has broken loose from the bone – splinting for four to six weeks may be necessary to ensure optimal healing of the fracture. Climbing can be resumed once imaging shows healing, when swelling has subsided, full mobility in the joint has been regained and there is no longer any pain. If, over time, however, the fracture doesn't heal properly, spot-drilling may be considered. This is a technique where small holes are drilled into the bone to allow blood vessels to grow in and contribute to fracture healing. Fortunately, this scenario rarely occurs; if symptoms are taken seriously early on, the finger will fully develop without future problems arising from such injuries. However, if left untreated for too long, the finger will suffer from permanent and irreversible damage, which will ultimately affect finger growth and development and may cause the young person to quit climbing altogether. Finger deformities will also make it difficult to engage in other fine motor activities later in life, emphasising how crucial it is that we take these injuries seriously!

DID YOU KNOW?

Growth plates are also referred to as epiphyseal plates and epiphyseal discs.

Since young climbers are at risk of developing these injuries until the growth plates have ossified, their climbing training during adolescence should be structured to reduce the use of the crimp grip and avoid campus board exercises. Caution should also be exercised when bouldering, particularly avoiding long sessions which focus on finger-intensive bouldering. Fortunately, climbing offers a wide range of possibilities for variation, and engaging in more technically demanding rather than physically demanding

climbing in combination with open-handed grip types will serve as good building blocks for developing skills and resilience in the long run. Additionally, there is plenty of physical training for the arms and upper body that does not negatively impact the fingers. Therefore, even with an injury requiring adjustments to finger load, one can emerge from this period as a stronger climber with improved technique.

REMEMBER:

- A growing skeleton is susceptible to growth plate injuries.
- The growth plates in the fingers fully fuse between the ages of 17 and 19.
- Adolescents going through a growth spurt are particularly prone to growth plate injuries, and most of them sustain this injury between the ages of 13 and 15.
- It is usually the PIP joints of the middle fingers that are affected.
- Injuries to growth plates are closely related to the number of climbing hours per week, use of the crimp grip, and specific training methods such as campus boarding and finger-intensive bouldering.

- If there is pain and swelling in the joint, one should take a break from climbing for a week. If symptoms do not improve with this rest period, it is advisable to seek medical attention and have an X-ray or MRI scan of the finger to assess the injury.
- Usually, a rest period of six to eight weeks is recommended before another MRI scan can determine if healing has occurred. If this rehab approach is followed, the prognosis is good, and the climber can expect to return to their previous level within three to four months. However, bear in mind that it may take longer than this, and young climbers and their coaches and parents should be aware of this type of injury throughout adolescence.



TRAINING, WEIGHT AND NUTRITION IN ADOLESCENCE

During puberty, both body height and body mass increase. Increased body mass leads to increased body weight, thus placing a higher load on the fingers and thereby increasing the injury risk. In girls, this period is estimated to take place between the ages of 11 and 14, and in boys between the ages of 12 and 16. Puberty often coincides with the time when many ambitious young climbers gradually increase their training load, and this combination is not without problems.

Dealing with an increase in body weight during puberty can be very challenging. Many will experience a decline in performance over a period of time, and may feel that their climbing peers are progressing faster than they are. It is particularly unfair for girls, as they don't experience the same muscle growth as boys during puberty and they also tend to gain more fat mass. Therefore, despite putting in the same effort in training, the results may not come as expected, and this obviously affects motivation. Knowledge of how the body develops throughout puberty can help both coaches and athletes adjust their training during certain periods; there are many areas for improvement and development in climbing even if progress in pure muscle strength does not always meet expectations.



REDs

REDs stands for **Relative Energy Deficiency in Sport**, and occurs when athletes consume less energy than they need. In weight-dependent sports like climbing we can influence the relationship between strength and body weight by becoming stronger or lighter. Reducing weight involves operating at an energy deficit – meaning that we provide the body with less energy than it requires. And, just like going bankrupt where we spend more money than we earn, operating at an energy deficit over time is unsustainable and can lead to serious health consequences.

The interrelationship between low energy availability (with or without disordered eating), menstrual dysfunction (amenorrhea) and low bone mineral density (osteoporosis) is referred to as the *female athlete triad*. The combination of these three components can lead to irreversible health consequences. In some cases, the focus on food and weight can also result in disordered eating or an eating disorder. This is a mental illness that causes both physical and psychological complications for health, and it is potentially fatal. While the female athlete triad only affects women, REDs can occur in both female and male athletes, manifesting as impaired bone health. Additionally, REDs can hinder development in young athletes, weaken the immune system,

increase fatigue, increase injury rates, reduce recovery capacity and diminish coordination, strength and concentration.

The use of weight reduction as a performance tool can cause severe physical and psychological injuries that may force individuals to quit climbing. Moreover, it is impossible to train enough – or well enough – over time if constantly operating at an energy deficit. Eventually, fitness levels will decline significantly, and it's often difficult to regain previous levels of performance through training alone. Many people lack the motivation required for this process and therefore they end up quitting climbing instead. It's extremely important to establish a healthy training environment, where young climbers are allowed to develop at their own pace and where nutrition and puberty are natural topics of discussion. This way we can reduce the risk of early performance-oriented goals, where weight reduction is used as a means along the way.

However, it is important to emphasise that REDs does not necessarily have to exist alongside an eating disorder or arise as a result of desired weight loss. It can also be related to the energy available for exercise. If you haven't eaten enough when you use more energy, your body won't be able to maintain all its functions.

So, even if you're not intentionally trying to lose weight or you see a weight reduction on the scales, it's possible that you still aren't getting enough energy. This is important to be aware of for younger athletes, who have high energy demands as they grow and often have busy schedules and may forget to eat along the way.

The Norwegian Olympic and Paralympic Committee and Confederation of Sports has a priority programme called *Healthy Sports*. It has listed characteristics that one can look out for which indicate a risk of developing REDs:

- disordered eating behaviour
- strong focus on perfectionism regarding food and exercise
- recurring infections and illnesses
- frequent injuries
- stress reactions/stress fractures in the skeleton
- menstrual disorders
- decreased libido

If you can identify with several of these points, it is important to ask yourself the following questions:

1. Am I exercising too much?
2. Am I eating enough/often enough?
3. Am I allowing enough time for rest and recovery?

This section started with a discussion of stress reactions in the growth plates of the finger joints, and ended by discussing growth, weight and nutrition. Growth plate injuries can occur due to a growing body being combined with increased training load alone, but it's important that we have an awareness of growth, weight and nutritional issues in young climbers. Almost all injuries are a result of multiple factors, and the most important thing we should ensure is that young climbers take care of their health and lay a solid foundation for climbing as a lifelong activity.

REDs

Relative Energy Deficiency in Sport

WHAT IS IT?



A condition of energy deficiency causing adverse effects on all bodily systems. It affects both male and female athletes who do not fuel adequately, whether intentionally or not.

RECOGNITION & NEXT STEPS

- Perfectionist tendencies
- Disordered/restricted eating
- Frequent injuries/niggles
- Illness
- Menstrual dysfunction
- Loss of sex drive

ARE YOU ...



- Overtraining?
- Fuelling adequately?
- Taking regular rest days?

SEEK MEDICAL HELP



- GP can rule out other conditions
- GP can refer to specialist services

WHY DOES IT MATTER?



1. Impaired growth and development
2. Impact on health and well-being
3. Adverse effect on performance

FOR FEMALE ATHLETES:

- Regular menstrual cycle is a barometer of hormone health.
- Not starting periods by age 16, or not having periods for >6 months requires medical investigation.
- Oral contraceptive pill (OCP) can mask problems without providing bone protection or addressing underlying cause.
- Using OCP for contraception needs to be an informed decision by the athlete.

BE AWARE THAT ...

20%

Prevalence of disordered eating is 20% higher among athletes compared to non-athletes.

Dudgeon, Emily, 'Relative energy deficiency in sport (RED-S): recognition and next steps', 2019, BMJ blog (22 April 2019), <https://blogs.bmj.com/bjbm/2019/04/22/relative-energy-deficiency-in-sport-red-s-recognition-and-next-steps> Reprinted with permission.

8 TRAINING RECOMMENDATIONS FOR YOUNG CLIMBERS:

1. Make training varied and enjoyable.
2. Focus on many moves rather than hard moves.
3. Spend most of the training time learning technical fundamentals and developing good movement patterns on the wall.
4. Train strength for the arms and upper body but pay careful attention to the training methods used for specific finger strength training. Focus on technical execution, simple training methods, low training volume and gradual progression.
5. Use basic strength and conditioning training for overall body strength, as well as balance, coordination and endurance training.
6. Eat healthy food, eat enough and eat regularly.
7. Pay attention to any symptoms in the fingers throughout puberty, but especially during periods of rapid increase in height and body mass.
8. BE PATIENT!



FLEXOR MUSCLES AND LUMBRICALS

'It really hurts when I load the fingertip. It hurts in the wrist and up into the forearm. But I can crimp without it hurting. I don't remember exactly what happened, but there was a boulder with pockets, my foot

popped and then there was a sharp pain in my forearm.'

In an open three-finger position, the PIP joint is stretched and the DIP joint is bent. Imagine hanging like this and slipping off a foothold. You would experience an acute increase in stress on the FDP, which could lead to a muscle and/or tendon strain injury.



We have two sets of muscles that flex the fingers, the flexor digitorum superficialis (FDS) and the flexor digitorum profundus (FDP). The former bends the middle (PIP) finger joint, and the latter bends the outermost (DIP) finger joint. In an open grip position (see [photo above](#)), the middle joint is extended while we bend at the outermost joint; this means that the FDP must bear most of the load in this grip position.

MECHANISM OF INJURY AND DIAGNOSIS

With a sudden increase in load – such as from a foot popping, your pinky finger slipping off while in a four-finger half crimp, or latching a hold in an open two- or three-finger position – you can sustain a strain injury to either the muscle, the tendon, or both. It is usually the ring finger that is affected, since for most people it is shorter than their index and middle fingers; while the ring finger remains straight at its PIP joint, the other two fingers still maintain some degree of flexion at their PIP joints. With a muscle strain injury, pain will be localised to the wrist and forearm, as this is where the muscle mass is located. A tendon strain injury can be a partial or a complete rupture; a total rupture of the FDP will leave you unable to flex the DIP joint. Fortunately, tendon injuries are far less common compared with muscle injuries which occur relatively frequently. I say fortunately, because tendon ruptures require a longer healing time, and surgery in severe cases, whereas muscle injuries heal relatively quickly. A clinical examination, supplemented by ultrasound examination if necessary, will provide insight regarding the severity of the injury, and this is important for determining the best course of action.

Muscle injuries are graded based on the degree of tissue damage, referred to as a rupture or tear. There is currently no literature specifically grading strain injuries in the FDP, but clinical experience suggests that most injuries are minor muscle ruptures. This corresponds to grade 1 in the general classification of muscle injuries.

CLASSIFICATION OF MUSCLE INJURIES

There are various grading tables for muscle injuries, but Pollock and colleagues' table from 2014 (below) is the most recent and is also well documented. Each of the numerical gradings can be further divided into a, b or c, depending on where the injury occurs within the muscle-tendon unit, and this will have implications for rehabilitation time: a muscle injury involving parts of the tendon (type c) will take longer to heal than injuries limited to the muscle belly and muscle-tendon (myotendinous) junction (MTJ) (types a and b).

British Athletics muscle injury classification	Grade 1 (small tear)	Increasing severity of symptom, signs and radiological disturbance
	Grade 2 (moderate tear)	
	Grade 3 (extensive tear)	
	Grade 4 (complete tear)	

MANAGEMENT

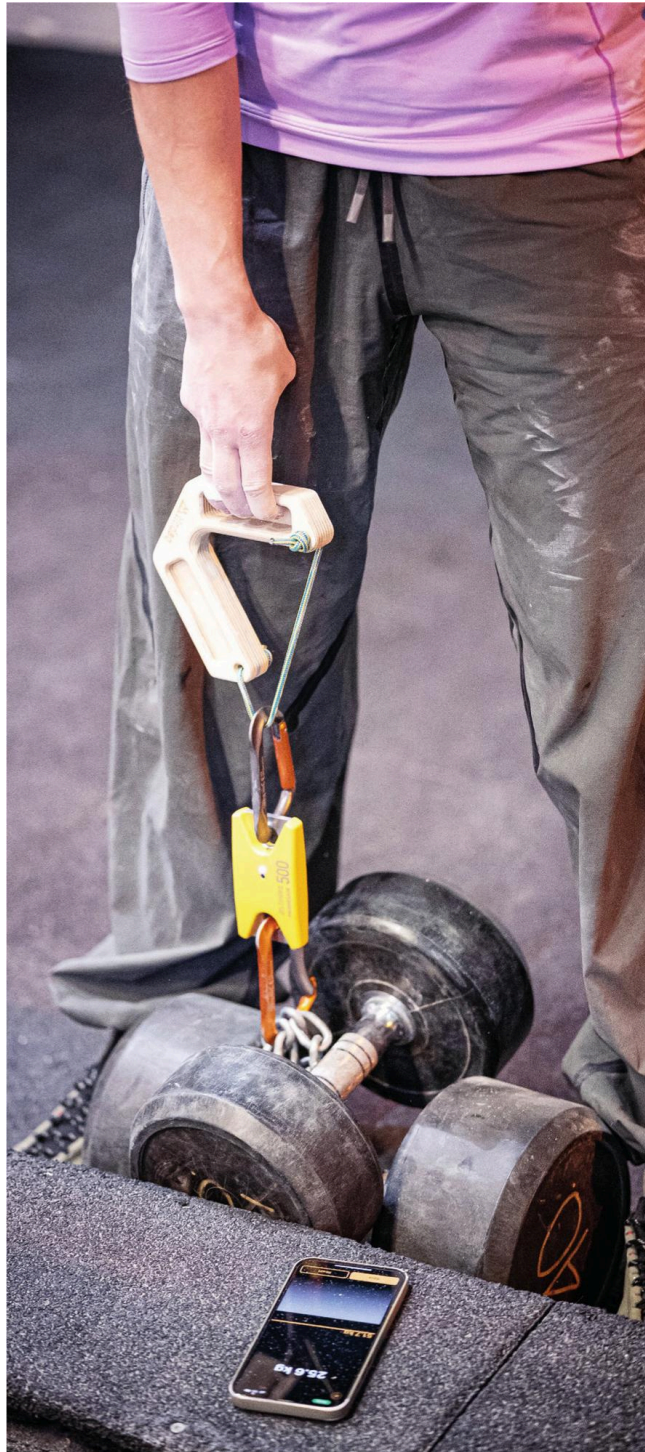
The positive thing about these injuries is that you can keep training and climbing, as long as you hang and climb with four fingers. When you place the pinky finger on a hold, you must bend the PIP joints in the other three fingers. By doing this, you engage the FDS, in addition to protecting the FDP from the direct strain it receives in the open three-finger grip position. This allows you to start loading the injured muscle early on, which is important for ensuring optimal healing and maintaining your existing physical capacity. The most controllable way to start the rehabilitation is by using a fingerboard or a grip tool, using a four-finger half-crimp position. This gives you control over the grip position and the number of kilos added, allowing you to adjust the load based on your symptom response. It is also important to use an open grip position to specifically target the

FDP, and by using the same tools, you can easily reduce the weight from a four-finger half crimp and use three, two or one fingers in an open grip position. The progression ladder is similar to that after pulley injuries (see pages [49-51](#)), but this time in open grip positions.

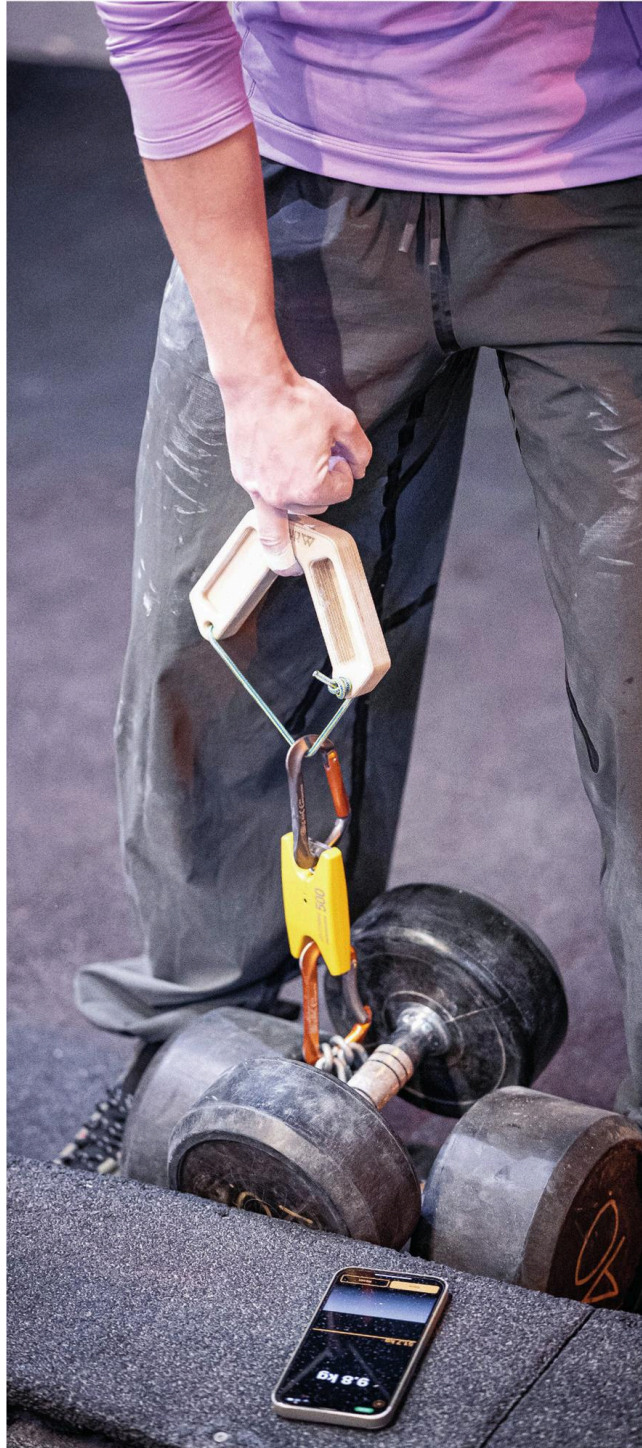
As with all muscle injuries, there is a risk of re-injury if you progress too fast; therefore, it's important to have control over grip position, load and movement tempo during rehabilitation. Based on experience, full recovery can be expected within six to eight weeks if recurring strains are avoided during this period.



This photo shows a test of the deep flexor muscle (FDP) of the ring finger. With the other fingers straight, the fingertip is pressed against resistance while the PIP joint is stretched. Pain in the forearm and wrist indicates a strain to the FDP.



After an injury, it's important to load with an open grip to give your muscles a specific stretch. By using a grip tool and a load cell, you can control the load and the speed of the movement, and pull as hard as you feel comfortable.



This grip position can be used to train both the FDP and the lumbricals.

LUMBRICAL SHIFT SYNDROME

The lumbrical muscles are located in the palm of the hand, and are attached to the flexor tendons. When we climb on pockets, the neighbouring fingers to the ones being used will automatically flex to generate more force. This results in a 'shift', where one flexor tendon shortens while the neighbouring tendon lengthens, causing the lumbrical muscle that is attached to both tendons to stretch. The majority of injuries are classified as grade 1 or 2 muscle injuries and are initially diagnosed through clinical examination. In cases where one suspects more extensive muscle/tendon damage, an ultrasound examination can be performed to supplement the diagnosis. This injury can occur independently, but it is often seen in conjunction with injuries to the flexor muscles, as described on the previous pages. The pain is localised in the palm, but as with injuries to the flexor muscles, it may be possible to climb without symptoms using four fingers in a half-crimp or full-crimp position.

Rehabilitation follows the same progression as for injuries to the flexor muscles, but it may be appropriate to tape the fingers to reduce the risk of re-injury. During acute phases, buddy taping is recommended, while more extensive injuries might require a taping technique which reduces how much the neighbouring fingers can bend.



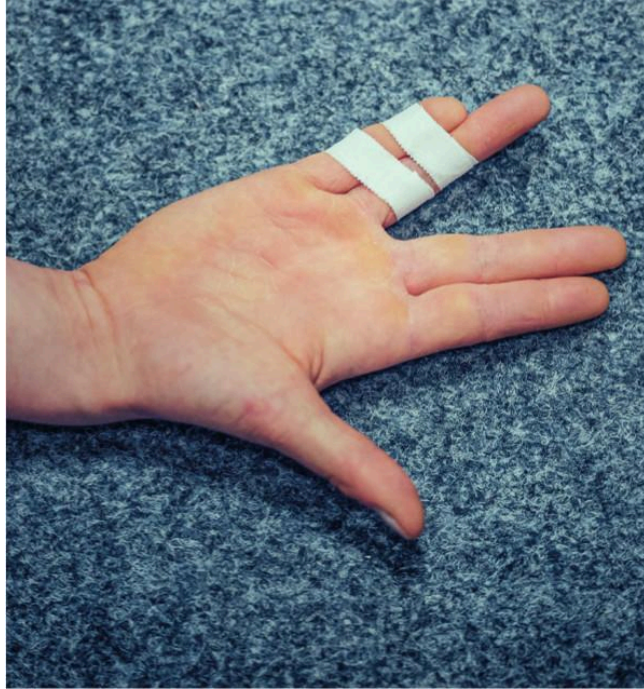
The 3rd and 4th lumbricals are most susceptible to injury because they are attached to two flexor tendons. If the ring finger tendon is stretched at the same time as the pinky finger tendon is shortened, the 4th lumbrical is stretched in two directions and can be damaged if the forces are great enough.



STRESS TEST FOR THE LUMBRICALS. Press the tip of the finger against resistance with the PIP joint extended and the other fingers flexed towards the palm. Pain in the palm over the relevant lumbrical muscles indicates a strain injury.



This method of taping allows more freedom of movement, but also limits the amount of flexion and extension of the three fingers in relation to each other. When recovering from a major lumbrical injury, this taping can be used to reduce the risk of recurrence when climbing.



Buddy taping prevents the pinky finger from bending inwards, thus preventing a lumbrical shift between the ring and pinky fingers.

DUPUYTREN'S CONTRACTURE

This is not actually an injury, but a condition that can be influenced and caused by several factors – such as genetics, diabetes and manual labour – and is traditionally observed in men aged between 40 and 60. However, climbers also experience it at a younger age, probably due to the stress we put on the connective tissue in the palms of our hands.

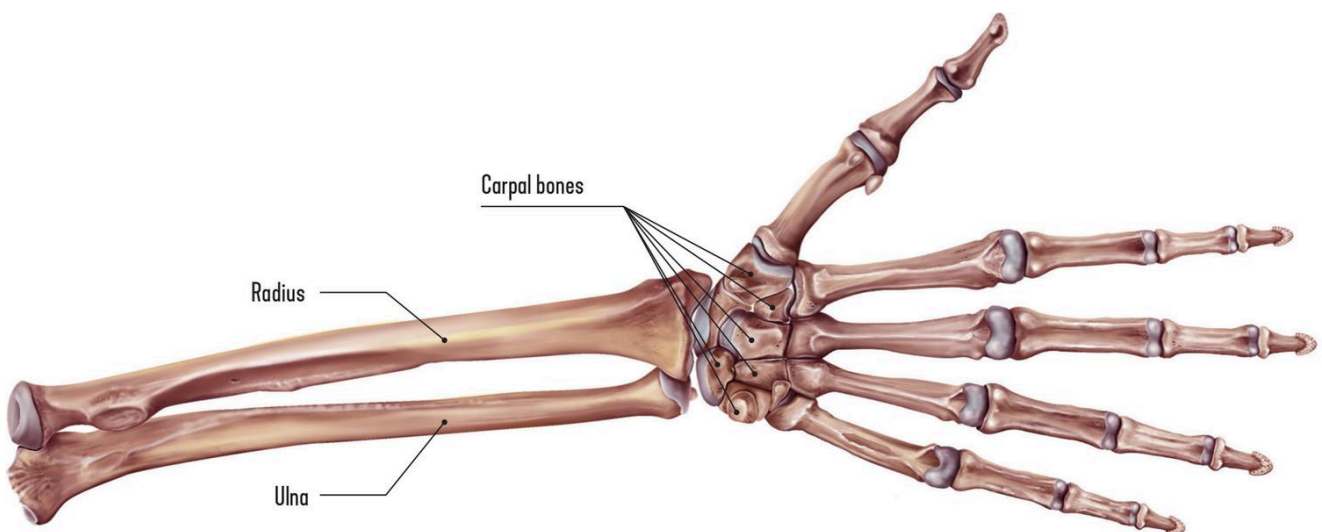
This connective tissue – called the palmar aponeurosis – can then contract so that it feels like a hard lump in the palm of the hand, usually in relation to the ring or little fingers. For most people, this is the only symptom, but in more severe cases, this contracture will cause the ring finger to remain flexed at its base (MCP) joint, and surgery may be considered if there's a fixed and significant contracture of the finger. It seems that there is little we can do to prevent or treat this condition beyond maintaining a healthy lifestyle and trying to maintain mobility with stretching exercises if symptoms do occur. However, it is a harmless condition that should not require taking a break from climbing unless surgery becomes necessary.



DUPUYTREN'S CONTRACTURE. In this case, the misalignment in the ring finger became significant enough that surgery was necessary.

HANDS AND WRISTS

The wrist is formed by the forearm bones – radius and ulna – and the carpal bones of the hand. The ulnar side of the wrist refers to the pinky finger side, while the radial side refers to the thumb side.



Climbing-related problems in the wrist range from acute and severe injuries, to load-related issues which will only require a change in loading pattern. To distinguish between these, we must first be aware of where in the wrist it hurts and how the problem has arisen.

REPETITIVE STRAIN INJURIES IN THE WRIST

Symptoms that arise gradually are usually related to irritation or inflammation (synovitis) in the wrist, or bone marrow oedema in the forearm and/or carpal bones. Bone marrow oedema can be compared to a bruise on a bone, caused by compression of the joint surfaces. Synovitis is the inflammation of the synovial membrane due to high stress within the joints (see pages [52–53](#)).

Pain is most often located on the ulnar side of the wrist, which is not surprising considering how we flex our wrists towards the ulnar side when we grip. In some cases, such as ‘wrapping’ a volume (see [overleaf](#)) or twisting into an undercut, this flexion combines with a rotation of the wrist, stressing the joint to a high degree. The same stress occurs during a mantelshelf, when we place much of our body weight on to our palms and push up.



Triangular volumes and large holds invite you to ‘wrap’ your hands around the edge. As the pictures show, this exposes the wrist to both rotation and lateral flexion, while at the same time providing a firm grip.



Twisting into an undercut and then moving out of that position requires a combination of wrist extension and lateral flexion towards the pinky finger side.

Bone marrow oedema can be detected by MRI, while synovitis presents as increased fluid on ultrasound images of the wrist. However, imaging diagnostics are not necessary if these problems were not triggered by an acute trauma as it's unlikely that the course of rehabilitation will change because of any potential findings on imaging.

The rehabilitation approach involves adjusting loading patterns and continuing training without a worsening of symptoms afterwards. Adjusting loading patterns means being selective about grip positions and movements. If mantels cause pain, it's okay to avoid boulders or routes with difficult mantels for now. If a project involves using a sidepull or an undercut, and wrist pain is felt after a few attempts, instead work on other parts of the project or take a short break from it. Learning to train on a spray wall allows you to easily vary hold/grip type, wall angle and move length, enabling quality training without worsening symptoms. By using smaller holds, doing shorter moves, improving foot placements and using less steep wall angles, you have many options to help you to adapt the load on your wrists while still having great climbing sessions.

ACUTE WRIST INJURIES

Acute injuries, however, resulting from a fall on a hand or a sudden incident during a move, should be further investigated. There are several stabilising structures, especially on the ulnar side of the wrist, that can be damaged, leading to both structural instability in the wrist and recurrent synovitis. The most well known of these structures is the *triangular fibrocartilage complex* (TFCC), which is attached to the ulna and the carpal bones. This complex binds together the bones of the forearm and stabilises the wrist on its ulnar side. The most common mechanism of injury is falling on to an outstretched arm while using the hand for support. More relevant to us climbers is the stress this complex undergoes when we angle our wrist towards the pinky side of the wrist and rotate our forearm – again, think how the wrist is angled in an undercut position. If you experience an acute incident in such a gripping position, followed by pain and a feeling of instability and/or clicking on the ulnar side of your wrist, I recommend that you have it further evaluated. This can be done through a clinical examination, and be supplemented with an MRI scan.

Some types of TFCC injuries require surgical intervention, but, generally speaking, both acute and load-related TFCC injuries are treated with load management and training. The preferred approach is to avoid positions that provoke symptoms – which usually involve flexing towards the pinky side with or without rotation – while simultaneously working on regaining strength and mobility. However, you must expect symptoms to persist throughout this process, which can last for several months, and for improvement to be gradual and slow. To help stabilise the wrist during rehabilitation, you can use tape or orthoses to increase stability. In cases of recurrent synovitis, cortisone injections may be considered as an option to reduce inflammation and facilitate further rehabilitation.

Being aware that certain grip positions make the wrist more exposed to overuse and injury is the most important preventive measure you can take. You can then consider whether it is necessary to make that famous last attempt, or you can try adjusting your body position so that the wrist is in a less vulnerable position. Through systematic strength training, the wrist will be able to tolerate higher forces, and you will be able to maintain a more favourable joint position in the wrist while climbing. Specifically strengthening the wrist appears central to a reduction in the risk of injury.

On the following pages are suggestions for exercises for the wrist, which can be done both as rehabilitation and as strength training. After all, rehabilitation is simply training while injured, and you can easily control the load based on symptom development during and after training.

If the stabilising structures on the ulnar side of the wrist are injured, it may be advisable to use a brace or orthosis. This can provide a sense of improved stability, and give you confidence to expose the wrist to heavier loads and more demanding movements.



A simple technique to support the wrist on the pinky finger side is to apply circular tape around the wrist.



An orthosis (called a WristWidget®) specifically designed to support the TFCC.

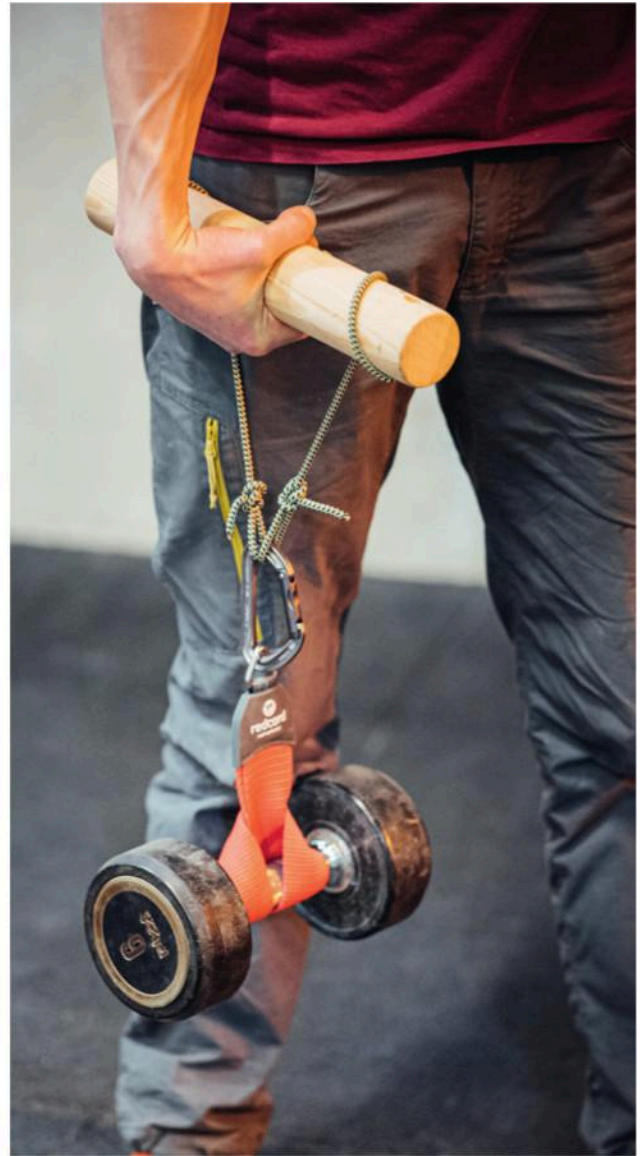
EXERCISES



PRACTISE HANGING FROM SLOPERS. As with deadhanging on edges, you can easily control the load by adding to or subtracting from your body weight and adjusting the hang time. To warm up, you can hang for 7–10 seconds, rest for 5 seconds and repeat 5–10 times. For strength training, follow the same progression ladder as for deadhangs on edges (see pages [49–51](#)).



Overcoming isometrics can also be used as a strength training exercise for wrist extension. Use a dumb-bell and place a wide sling over the back of your hand. Push your hand up as hard as you can and hold for 5 seconds. Rest for 1 minute and repeat for 5 sets. If it's painful, you can reduce the force and increase the holding time to 10–15 seconds.



To train wrist flexion in an open grip position, the wrist wrench is a good exercise. Use a wide cylinder (>5cm in diameter) and flex your wrist inwards. Repeat 6–8 times for 3 sets.



PICK-UP WITH PINCH BLOCK. Start with a weight that you can hold for no more than 10 seconds. Repeat 5 sets with 1–2 minutes' rest in between. Gradually increase the weight and reduce the holding time to 5 seconds per lift.



ULNAR FLEXION. In some cases you will want to load into ulnar flexion to strengthen this wrist position. Use a dumb-bell with a disc at one end and lift the weight as shown. Perform 6–8 repetitions for 3 sets.

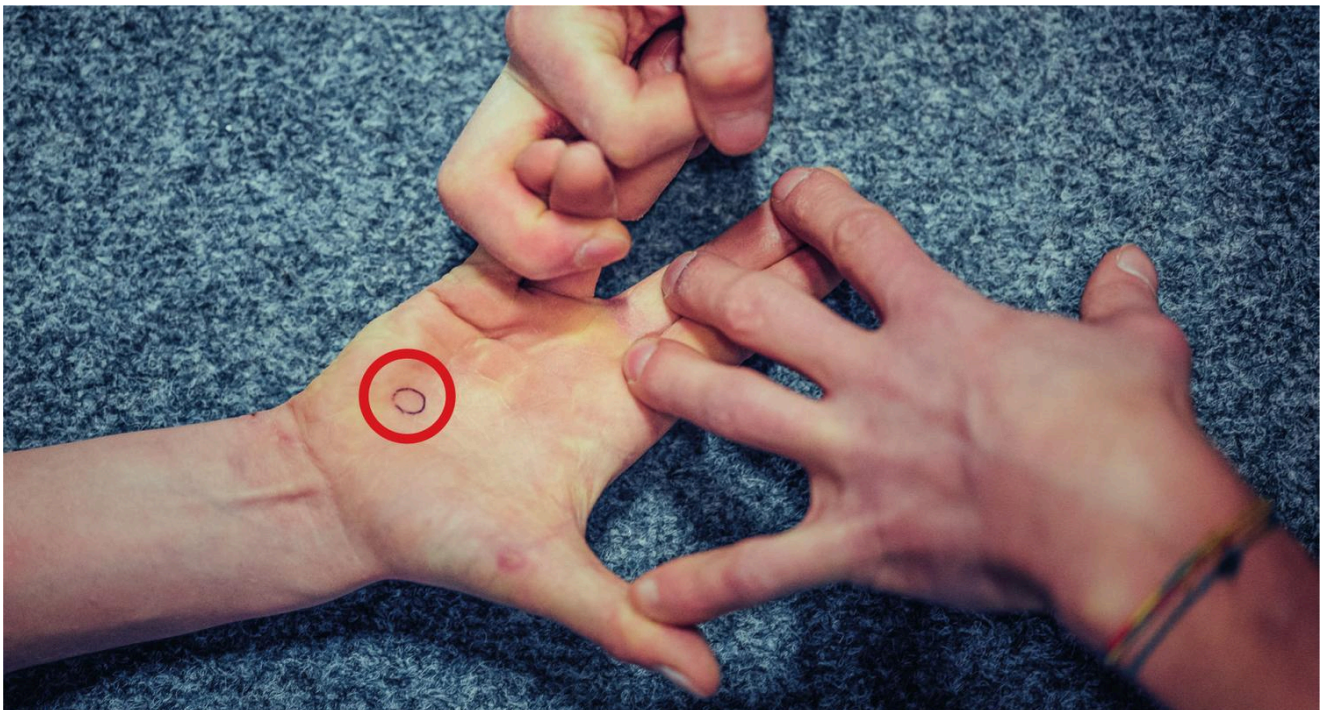


By putting weight on the wrist in small doses, you can gradually regain weight-bearing capacity. Start with your wrist in a neutral position (figure 1) by using your knuckles, then move to weight bearing on your palms (figure 2). Alternate between these positions as part of your warm-up, so that you have exercised your wrist for a total of 3–5 minutes per session.

HOOK OF HAMATE FRACTURE

Yes, I know, this was supposed to be a book about *common* injuries, and this injury is far from common! But it is very specific to climbing, and it is easy to overlook. I have missed it myself, and, unfortunately, it led to a longer rehabilitation than necessary for the unlucky climber. So, consider this an attempt to make up for that oversight, so that this injury is on your radar should it happen to you.

One of the carpal bones is called the *os hamatum*, and it has a hook that stabilises the flexor tendons of the ring and pinky fingers. The English name for this is the *hook of hamate*, and although the most common mechanism of injury is a direct trauma to the inside of the hand, the hook can actually break off from the rest of the bone through intense pressure from these two tendons. This occurs when you flex your wrist towards the pinky side and pull hard with the ring and pinky fingers. In climbing, typical examples are when using undercuts and sidepulls. The injury usually occurs acutely but can also result from repetitive stress over time. The pain is felt deep in the palm, usually a little closer to the pinky side than the thumb side. If there is tenderness when pressure is applied to the area, and a recognisable pain when the ring and pinky fingers are loaded, at least suspect this injury and request an MRI or CT scan. In the case of a fracture, the hand should be immobilised with a cast or support until it can be seen that the hook has fused firmly to the rest of the bone. This should be followed by a rehabilitation period, gradually increasing load while being aware of grip positions. Done properly, you will end up with a fully healed fracture that can withstand higher forces than before the injury. Failed healing might result in a dislocated fracture – meaning that the loose hook does not reattach itself to the rest of bone. In such cases, surgery becomes necessary, but if the injury is caught early on, few will need surgical intervention.

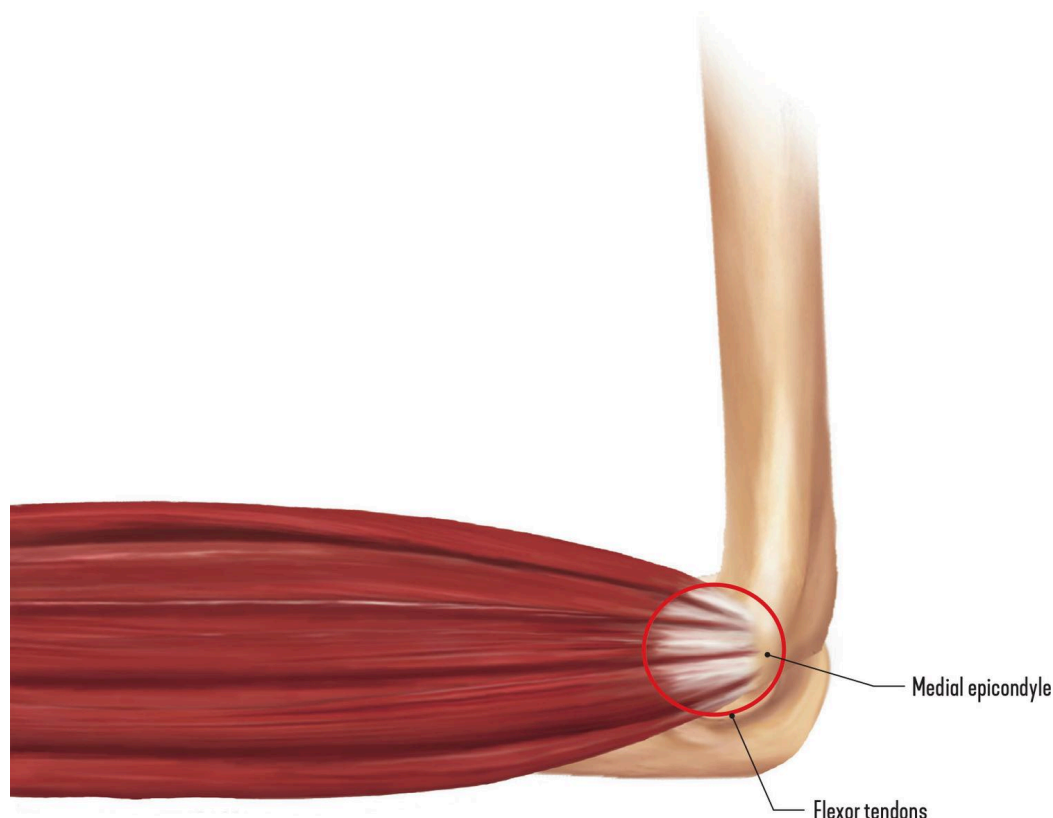


Hook of hamate tension test. Bend your wrist slightly upwards and towards your pinky finger. Then pull hard with your ring and little fingers, putting stress on the hook of hamate, which stabilises the flexor tendons of these two fingers. If there is pain in the area highlighted in the photo above, the hook may be damaged and you should seek a referral for an MRI or CT scan.



High stress in undercuts, where the wrist is angled towards the pinky finger side, is the most common mechanism of injury.

ELBOWS AND FOREARMS



Injuries and pain related to the elbows and forearms are mostly caused by excessive stress over time. However, the elbow is also prone to acute injuries from falls, especially during bouldering. Acute traumatic elbow injuries should always be evaluated by a healthcare professional to detect possible fractures and ligament damage; these injuries are not covered here. This section describes load-related issues related to the forearm muscles and their tendon origins on the inside and outside of the elbow, and pain in the middle aspect of the elbow.

MEDIAL EPICONDYLALGIA – AKA ‘GOLFER’S ELBOW’

‘It hurts when I press here, on the knuckle on the inside of my elbow. It’s painful when I warm up, but climbing can be fine. I notice that it gets worse after climbing, and then it hurts if I slice bread or twist bottle caps.’

The ‘knuckle’ referred to here is the bony protrusion on the inside – medial side – of the elbow. The knuckle is called the medial epicondyle, so medial epicondylalgia simply refers to pain in this area. The condition is also known as ‘golfer’s elbow’, although a quick Google search shows that it is now also referred to as ‘climber’s elbow’; however, since this name is reserved for another condition (see page [96](#)), we will stick with golfer’s elbow or medial epicondylalgia.

The common tendon for most of the muscles that flex the fingers and wrist and rotate the forearm so that the palm faces downwards (pronation), originates from the medial epicondyle, and the pain primarily relates to reactions in this tendon. The tendon can react due to excessive strain within a short period, but it is also subject to compression forces where it spreads out over the knuckle – particularly when we fully flex our elbows when locking off. Imagine what happens during a pull-up: we grip a bar with our palms facing away from us (a pronated position), using both the flexor muscles of the fingers and wrist and the pronator teres muscle while pulling up into a fully flexed elbow position. If we combine these forces of tension and compression, along with numerous repetitions, the tendon can react and cause pain. Similarly, a high climbing volume on steep or vertical terrain can cause excessive strain, leading to a negative reaction in the tendon.

Although the reaction is a result of doing too much, the solution is not to avoid loading the tendon but rather to break the load down into different components and train them separately.

It is important to emphasise that there is no perfect protocol or training method; instead, we must rely on some basic principles of loading and adaptation in tendon tissue. This provides us with ample opportunities to